



YEAR 13 PROJECT · 2025

Self-Landing Rockets on a Budget

How I designed, built and ground-tested a thrust-vector-controlled model rocket with hobby parts and a 3D printer.



“To what extent do modern technologies enable the construction of a self-landing rocket by individuals without a billion-dollar budget?”

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Rockets that land themselves

For most of the history of spaceflight, rockets were used once and thrown away. Their stages burned up in the atmosphere, sank into remote “rocket graveyards” in the ocean, or stayed in orbit as debris. Imagine scrapping a plane after every flight across the Pacific: that is roughly what launching used to cost.

In 2015 that changed. Blue Origin’s New Shepard landed vertically after reaching space, and soon after, SpaceX landed the first stage of a Falcon 9 that had just delivered a payload to orbit. Watching those boosters touch down, sometimes several at once, is what got me hooked.

Reusability is now how the industry works. Rocket Lab in New Zealand is building its reusable Neutron rocket for exactly this reason, which I saw first-hand when I toured their Auckland facility for this project.

That made me curious whether the same ideas could be scaled down to a student budget, using tools I actually had: CAD software, a 3D printer, hobby servos and small computers.

Why it only became possible recently

Four things had to mature at once. **Computers** became fast enough to make millisecond-level corrections on the way down. **Manufacturing** changed: 3D printing and CNC machining made fast prototyping and quick refurbishment affordable, and Rocket Lab even 3D-prints parts of its Rutherford engines. **Engines** learned to restart, throttle deeply and gimbal to steer. And **commercial competition** since the 1980s gave private companies room to take risks that Cold War agencies, chasing reliability and prestige, never would.

\$20,000 → \$2,700

Cost per kilogram to orbit, from the Space Shuttle era to today’s reusable Falcon 9. A drop of about 86 %.¹

>\$1.5 billion

Cost of a single Space Shuttle launch. Its partial reusability needed so much refurbishment that it ended up dearer than expendable rockets.²

~400,000×

More additions per second in Falcon 9’s flight computer (about 4.8 billion) than in Saturn V’s (about 12,000).³

Why landing is so hard

1. Slowing down with the engine

Rockets have no brakes. A returning booster must burn its engine to slow from very high speed to exactly zero at ground level, without running out of fuel or shutting down too early. That needs engines that can restart and throttle, which early engines could not do.

2. Staying upright

Wind, changing air pressure and turbulence all try to tip the rocket over. If it tilts too far it can break apart, so the flight computer has to correct its attitude constantly, every few milliseconds, using a gimbaled engine, grid fins and small cold-gas thrusters.

3. Hitting the target

The rocket has to touch down on a pad or a drone ship a few metres across, coming from orbital speed. That needs precise navigation as well as control.

Thrust vector control, in one sentence.

Tilting the engine a few degrees moves the line the thrust pushes along away from the rocket's centre of gravity, which creates a turning force that rights the rocket. At zero tilt the thrust passes straight through the centre of gravity and there is no turning force at all.

Shrinking it down

The good news for a student is that the hardware has shrunk and become cheap:

- **Microcontrollers** like the Teensy 4.1 can run control loops fast enough to stabilise a small rocket.
- **Sensors** that used to be military-grade, like IMUs (gyroscope and accelerometer) and GPS, now cost a few dollars.
- **Precise hobby servos** under US\$50 can move a gimbal.
- **Free design and simulation tools** such as Fusion 360, FreeCAD and OpenRocket.
- **Open knowledge** from forums, GitHub and other builders' projects.

The rocket



The finished rocket on its four landing legs. Nearly everything you see is 3D-printed.

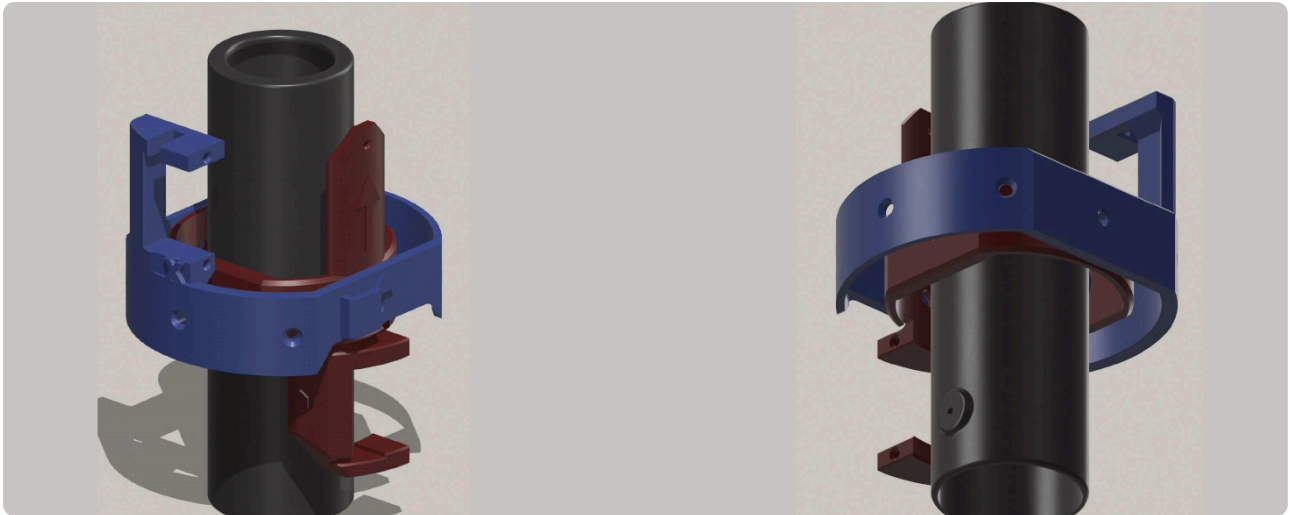
The rocket is built around the two systems that make self-landing possible: steering and thrust control. Almost all of the structure was designed in CAD and printed on a Bambu Lab X1 Carbon, in carbon-fibre-reinforced nylon for strong parts and lightweight PLA Aero for the shell.

What's inside

- **Shell and nose cone**, printed in four body sections plus an ogive nose cone.
- **TVC gimbal** that tilts the motor on two axes to steer.
- **Throttle mechanism**, an experiment for controlling the thrust of a solid motor.
- **Flight computer**: a Teensy 4.1 with an IMU, barometer, GPS, ultrasonic rangefinder and temperature sensors.
- **LoRa radio** for live telemetry, received by a Raspberry Pi Zero W ground station that logs the data.
- **Power** from 3S LiPo batteries in parallel, regulated to 8.4 V for the servos and 5 V for the electronics.
- **Four A-frame landing legs**.

Being able to print a part was the biggest advantage. When a servo mount snapped, I could fix the CAD and have a new one within hours, the same rapid-iteration approach I saw at Rocket Lab.

Thrust vector control

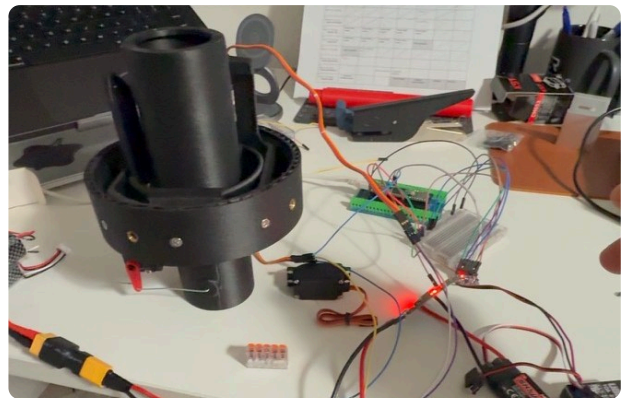


My TVC gimbal in CAD. The motor tube sits in an inner ring (red) that pivots inside an outer ring (blue). Each ring is driven by its own servo, which gives two axes of steering.

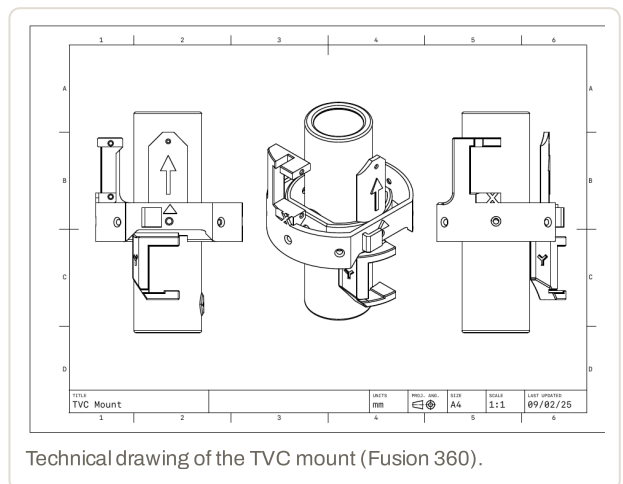
The TVC mount is a two-axis gimbal. Two KST CM653 servos push the rings, tilting the motor a few degrees in any direction. The flight computer reads the IMU, works out how far the rocket is leaning, and moves the servos to push it back upright.

I designed the mount to be as light as possible while still taking the motor's force. That's the same trade-off SpaceX faces with its engine gimbals, at a much smaller scale.

The first versions jammed under load. I blamed the material at first, but the real problem was my own tolerances in CAD, which were too tight once the parts were printed.

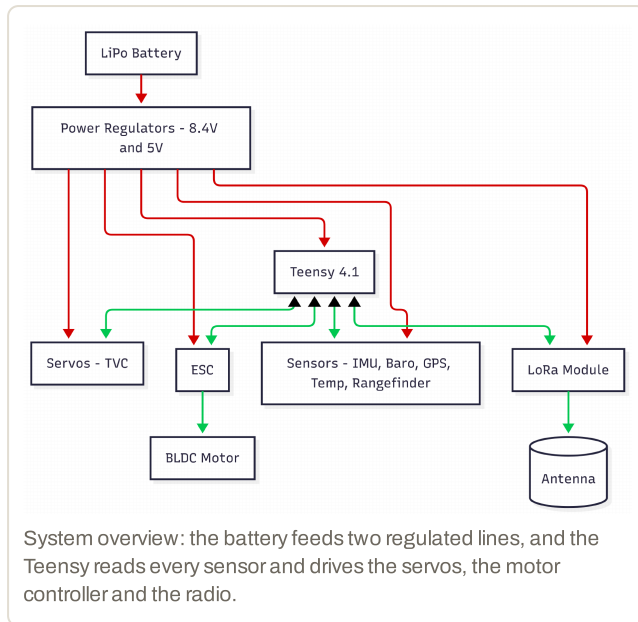


The printed gimbal on my desk during bench testing, wired to the servos and controller.



Technical drawing of the TVC mount (Fusion 360).

Flight computer and electronics



The Teensy 4.1 runs the control loop. It reads the sensors, decides how to correct, and drives the servos, all with very low latency, which matters when a rocket can tip over in milliseconds.

- **IMU** for orientation: which way and how fast it's tipping.
- **Barometer** for altitude.
- **Ultrasonic rangefinder** for accurate height close to the ground.
- **GPS** to find its way back to the pad.
- **Temperature sensors and battery monitors** for health checks.
- **LoRa radio** streaming telemetry to the ground station.

What it cost

PART	MODEL	QTY	TOTAL
Microcontroller	Teensy 4.1	1	\$63.56
Servos	KST CM653	4	\$176.60
IMU	MPU9250	2	\$11.08
Barometer	BMP388	2	\$11.28
GPS module	GY-NEO8MV2	2	\$23.74
Temperature sensors	DS18B20	4	\$14.68
Battery monitors	INA3221	2	\$3.40
LoRa telemetry	SX1278	2	\$12.16
Batteries	3S LiPo	3	\$71.94
Power distribution board	Matek FCHUB-12S	1	\$49.00
Servo power supply (UBEC)	Hobbywing 10A 8.4V	1	\$29.04
Test motor + speed controller	Dualsky ECO2312C + Hobbywing Skywalker 40A	2	\$56.30
Ultrasonic rangefinder	HC-SR04	1	\$3.54
Total electronics			\$526.32

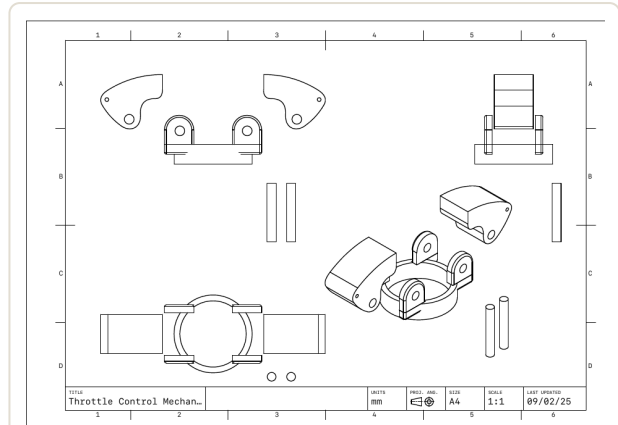
The whole electronic system cost about the same as a mid-range phone. That's a fair answer to the “without a billion-dollar budget” part of my question.

Throttling a solid motor

Landing needs throttle control, and solid rocket motors can't be throttled: once they're lit, they burn at their designed rate until they run out.

My workaround was a mechanism that pushes small ceramic “shovels” into the exhaust to partly block it, which reduces the useful thrust. Their mounts are 3D-printed in several parts and reinforced with metal plates against the heat. The shovels themselves would need to be CNC-machined.

It felt ridiculous at times, pushing ceramic paddles into a jet of exhaust. But working it out made me understand how professionals solve the same problem in liquid engines, where throttling is built in. It stayed a prototype: without a motor, I could only test that the electronics and code controlling it worked.

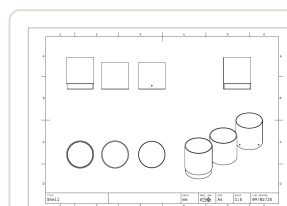


Technical drawing of the throttle-control prototype.

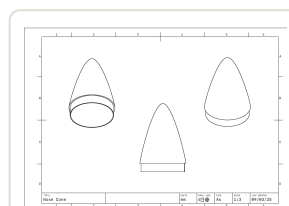
Shell and nose cone



The shell assembled (left) and split into its printed sections (right).



Body sections.



Nose cone.

Testing without a launch

Why no launch? I originally planned to make my own solid motor powerful enough to lift the rocket. The chemicals needed are tightly controlled in New Zealand, so a live flight wasn't possible. Real aerospace engineering runs into the same kind of gap, between what's possible in theory and what's allowed and safe, so I adapted the project to fit.

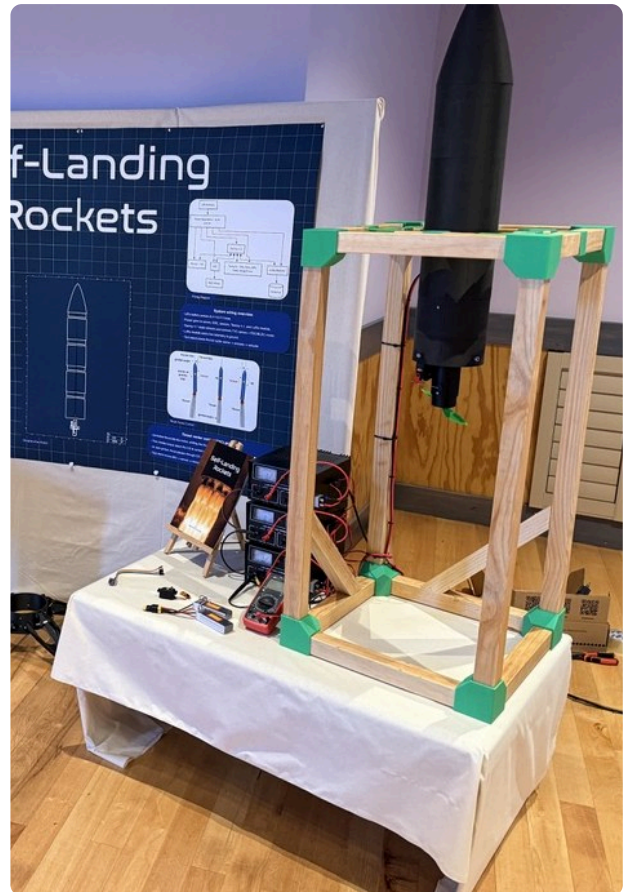
Instead of flying, I tested every system on the ground:

- **A static test stand:** a timber frame with printed joints that holds the rocket vertically by its middle, so the tail can move freely.
- **An electric propeller rig** in place of the rocket motor. A brushless motor provides real, controllable thrust, so the TVC system can be tested over and over without burning any fuel.

What the tests showed

The TVC mechanism works mechanically: the servos move the gimbal fast and accurately enough to correct the rocket. The hard part is software. Tuning the PID controller is difficult, because small errors compound very quickly and a servo that reacts a fraction too late makes the whole system unstable.

The throttle hardware couldn't be tested without a motor, but I connected the electric test motor to the same circuit, which proves the electronics and code for throttle control work.



Presenting the project: the rocket mounted in its static test stand, with the bench electronics in front and my poster behind.

What I learned

My first picture of this project, a rocket lifting off and gently touching down again, was a bit naive. When the propellant restrictions ruled out a launch, it felt like failure at first. Over time I realised the detour was the real lesson: engineering almost always hits barriers, and progress depends on how you respond.

Technical

- Sensor fusion, PID control, and why latency matters when a rocket can tip in milliseconds.
- Designing for printing: tolerances, infill, carbon-fibre filaments, and the trade-off between speed and strength.
- That CAD designs often look better on screen than they behave on the workbench.

Mindset

- How to pivot. Switching to a static stand with an electric motor saved the project.
- Patience through a constant cycle of optimism and small failures.
- Each broken part points at a real weakness, and that's worth more than a polished tutorial.

What I'd do differently

I spent too long early on researching motor chemistry I never got to use. If I started again I'd build small single-axis test rigs first, before the full two-axis gimbal, to get clearer data with less frustration.

The answer to my question

Modern technology gets an individual surprisingly far. Thrust vector control is achievable with hobby-grade parts, 3D printing, and around \$500 of electronics. Throttling a solid motor is far harder, and live flight is limited by law and safety rather than by technology. Self-landing rockets are possible at model scale, but only with compromises. Even without a launch, the project showed me how big the gap between theory and practice really is, and how to keep going when things don't work.

Thanks to Rocket Lab for the facility tour and their encouragement, and to my mentors, teachers and family for their support throughout the project.

Images are my own photos, CAD renders and drawings.

Sources. 1. American Institute of Aeronautics and Astronautics (2022), *Analysis: Bending the cost curve*, Aerospace America. 2. Jones, H. W. (2018), *The Recent Large Reduction in Space Launch Cost*, ICES. 3. Boyer, C. (2025), *The Saturn V Instrument Unit*; SmarterEveryDay (2019), *How did NASA Steer the Saturn V?*. The full research paper, with all references, is available on request.